ECON3700
ENVIRONMENTAL ECONOMICS

Environmental Analysis II

Benefit-Cost Analysis – Costs

References: Chapter 8, ISE Environmental Economics, 9th Edition

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Learning Objectives

1. Understand the role of costs in benefit-cost analysis.
2. Explain opportunity cost and its applications.
3. Explore with/without vs. before/after principles.
4. Identify issues in cost estimation at different scales.
5. Discuss distribution of costs in environmental programs.

Why focus on costs?

- Benefits often dominate analysis
- Ignoring costs = misleading policy conclusions

Examples: climate policy, waste cleanup, energy transition

The Myth of Cost-Benefit Analysis

Climate policy co-benefits: a review

Opportunity cost

- Opportunity cost = value of best alternative foregone
- Not just financial outlays
- Applies to time, labour, resources

Examples:

- Hiring unemployed workers > low opportunity cost
- Unpaid family labour > hidden but real
- Repurposing idle land > trade-offs

Environmental costs of environmental programs

- Fuel switching > new pollutants
- Cleaner tech > higher energy use
- Trade-offs across environmental media

No-cost policy changes

- Correcting price distortions
- Example: removing fossil fuel subsidies
- Benefits realised without direct spending

Enforcement costs

- Monitoring & compliance
- Administrative overhead
- Ongoing vs. one-off costs

Enforcement costs

Policy Area	Enforcement Activity	Examples of Enforcement Costs
Vehicle Emissions & EV Policy	Monitoring compliance with the **New Vehicle Efficiency Standard (NVES)**	Government systems to track emissions per km, audits of carmakers, penalties for breaches, IT platforms for reporting
Carbon Pricing & ACCUs	**Safeguard Mechanism** and carbon credit verification by the **Clean Energy Regulator**	Hiring auditors, compliance investigations, verifying carbon credit projects, preventing fraud (e.g., “phantom offsets”)
Waste & Recycling	**Container Deposit Scheme (CDS)** compliance; enforcing **plastic bans**	Inspections of refund depots, fraud prevention (bottle transport across states), auditing beverage suppliers, staff training
Energy Transition (Renewables)	Ensuring compliance with the **Renewable Energy Target (RET)**	Tracking Renewable Energy Certificates (RECs), IT systems for certificate trading, penalties for non-compliance by retailers
Land & Biodiversity Protection	State **EPA enforcement** of illegal land clearing, pollution, or habitat destruction	Site inspections, aerial surveillance, prosecutions, legal costs, environmental monitoring
Local Environmental Laws	Council enforcement of waste dumping, noise, and local pollution	Employing inspectors, issuing fines, investigation costs, public awareness campaigns

From facility to nation



Cost grow in complexity with scale

From facility to nation



- Direct expenditures
- Example: wastewater treatment plant
- Engineering-based costs

- Infrastructure impacts
- Public service demand
- Local economic ripple effects

Costs of Single Facilities

Table 8.1 shows costs for a representative 200-megawatt land-based wind turbine power plant.

Estimation of environmental costs not included?

- Cost of wildlife disruption
- Noise pollution
- Degradation of scenic values

Opportunity cost?

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TABLE 8.1 Typical Costs of a 105–Turbine, 200 Megawatt Wind Energy Plant

	Total ($1,000)	Annual ($1,000)
Capital Costs		
Site preparation	10,500	
Turbines	238,000	
Assembly	21,000	
Electrical infrastructure	31,000	
Finance	10,500	
Contingency	21,000	
Other	18,000	
Decommissioning	(not estimated)	
Total	350,000	28,000*
Annual Costs		
Operations		2,800
Maintenance		5,600
Land lease		1,600
Total		10,000
Environmental Costs	(not estimated)	
Capital costs per installed kw $1,750		

*Based on an interest rate of 5 percent and a life of 20 years.

Source: C. Mone, A. Smith, B. Maples, and M. Hand, *Cost of Wind Energy Review,* National Renewable Energy Laboratory., Technical Report NREL/TP-5000-63267, 2015.

From facility to nation

Industry

- Compliance costs across firms
- Competitiveness impacts
- Example: emissions trading scheme

National

- GDP impacts
- Employment shifts
- Structural changes

Costs of Regulating an Industry

EPA sets emission standards for significant sources of air pollutants.

Table 8.2 shows the costs of R&D for new devices.

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TABLE 8.2 Cost Estimates for Developing New Wood Burning Heaters to Meet Stricter Particulate Emission Rates

	Development Costs for "Best System of Emission Reduction" ($)	
R&D Engineering Costs		
Prototype design	$32,250	
Tooling	33,475	
Other	10,000	
Total		*$136,475*
EPA testing, certification	19,400	
Total		*155,875*
Annualized Capital Cost (6 yrs., 7%)		*32,702*
	Total Costs for All Wood Burning Stoves ($)	
	Typical Manufacturer	**All Manufacturers**
Wood stoves	$88,968	$3,020,000
Single burn heaters	352,506	870,000
Pellet stoves	51,179	1,520,000
Forced-air furnace	2,986,523	15,360,000
Hydronic system	828,513	24,880,000
Total		*45,660,000*

Source: USEPA, *Regulatory Impact Analysis for Residential Wood Heater Final Report,* EPA-452/R-15-001, 2015.

Cost of National Regulation

What is the macroeconomic cost burden of the environmental regulations a country imposes?

Bottom-up – looks at expenditure made throughout the economy by individual firms for pollution-control purposes.

Or

Top-down – looks at whether, and how much, pollution-control programs may change national economic performance.

Cost of National Regulation – The Environmental Industry

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Industry Segment	Revenue (Billion $)			
	1990	2002	2010	2017
Industry total	152.3	221.3	316.3	388.0
Analytical services[1]	1.5	1.3	1.8	1.9
Wastewater treatment works[2]	19.8	30.2	46.9	55.9
Solid-waste management[3]	26.1	42.7	52.4	57.9
Hazardous-waste management[4]	6.3	4.7	8.7	10.8
Remediation/industrial services	11.1	11.1	12.2	14.2
Consulting and engineering	12.5	18.7	26.2	28.9
Water equipment and chemicals	13.5	20.8	27.2	30.5
Instrument manufacturing	2.0	3.9	5.5	6.3
Air-pollution control equipment[5]	13.1	18.6	14.9	16.0
Waste-management equipment[6]	8.7	9.9	11.1	12.2
Process and prevention technology	0.4	1.4	1.9	2.1
Water utilities[7]	19.8	32.3	42.1	54.6
Resource recovery[8]	13.1	14.1	25.2	29.1
Clean energy systems and power[9]	4.3	11.5	40.1	NA

[1] Covers environmental laboratory testing and services.
[2] Mostly revenues collected by municipal entities.
[3] Covers such activities as collection, transportation, transfer stations, disposal, landfill ownership, and management for solid waste.
[4] Transportation and disposal of hazardous, medical, and nuclear waste.
[5] Includes stationary and mobile sources.
[6] Includes vehicles, containers, liners, processing, and remediation equipment.
[7] Revenues generated from the sale of water.
[8] Revenues generated from the sale of recovered metals, paper, plastic, and so on.
[9] Includes solar, wind, geothermal, and conservation devices.

Source: Environmental Business International, Inc., San Diego, CA, *Environmental Business Journal*, monthly (copyright) (as published in *Statistical Abstract of the United States: 1999*, 251, 2003, and 2012, 232); Department of Commerce, International Trade Administration, 2017, www.trade.gov/topmarkets/pdf/Environmental_Technologies_Executive_Summary.pdf.

Cost of National Regulation

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TABLE 8.3 Pollution Abatement & Control Expenditures (PAC) as a Percentage of Gross Domestic Product (GDP), Selected Organization for Economic Cooperation and Development (OECD) Countries, Latest Available Year

Country	PAC as a Percentage of GDP
Canada	2.2
France	1.1
Germany	1.3
Japan	1.2
Mexico	0.8
Korea	1.3
Brazil	1.1
United Kingdom	0.9
United States	1.2

Source: MARI Foundation, http://marifoundation.org/economic/2015.

Case study prompt

Co2 reduction in Australia – costs at each level?

Can you list them?

With/without principle *vs* before/after

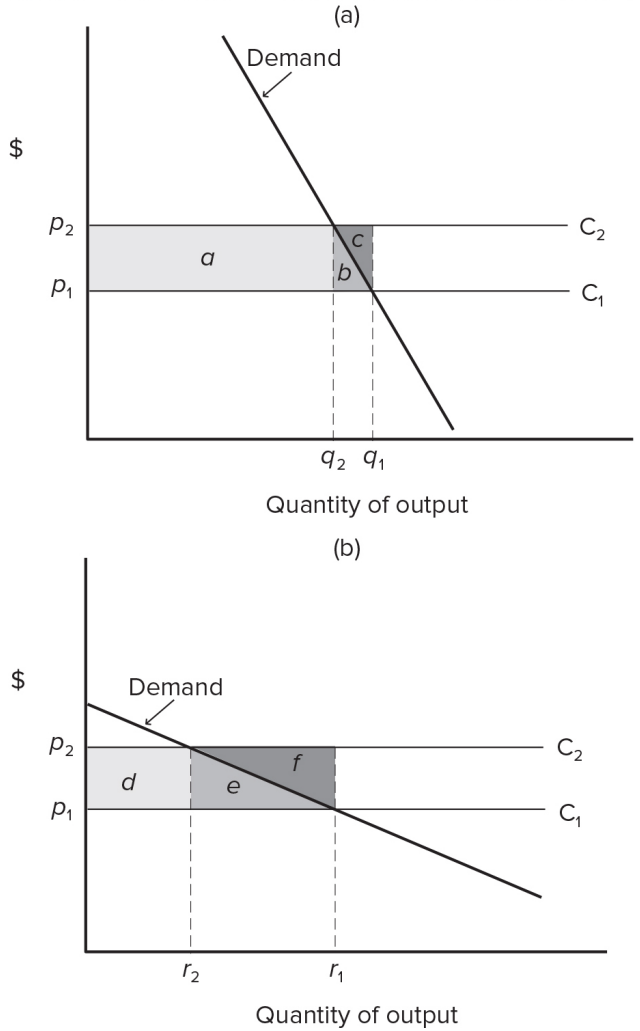
- Before/After can be misleading
- With/Without isolates project impact
- Example: river clean-up vs. natural recovery

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Before the regulation:	$100
In the future without the regulation:	$120
In the future with the regulation:	$150

The Effect of Output Adjustments on Costs??

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Cost increases > output prices?

Quantity demand?

Long-run cost estimation

Tech innovation > falling costs

- History shows that new technology often gets cheaper much faster than predicted.
- Example: solar PV

Structural shifts > unpredictable

- Entire sectors can change in ways that models fail to anticipate.
- Example: Energy markets – the shift from coal to renewables in Australia has been faster than expected

Forecasting is challenging

- Long time horizons (20–50 years for climate policy) introduce huge uncertainty.
- Interactions between policies - subsidies, carbon pricing, and regulations can interact in unpredictable ways.

Why distribution matters?

- Costs rarely shared equally
- Political fights about who pays
- Efficiency vs. equity